

Predicting Medicare Part D Drug Spending Patterns *and Identifying Prescriber Outliers*

A Data-Driven Framework for Pharmacy Benefit Management Formulary Optimization

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QM 640: Data Analytics Capstone | Walsh College
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Executive Summary

This project quantifies Medicare Part D's \$275.8B-a-year spending problem and identifies \$9.91 billion in defensible annual savings using a fully integrated CMS + Open Payments data pipeline and five complementary analytical models.

PROBLEM	APPROACH	FINDINGS	VALUE
Part D grew \$103.7B to \$275.8B, brand drugs are 85% of cost, no integrated decision framework exists for PBMs	Five integrated models on full CMS data (27M prescriber rows, 14.7M Open Payments records) plus BRFSS+ACS state demographics	OLS $R^2 = 0.92$, K-means $k=4$ clusters, $\eta^2 = 0.139$ specialty effect, Open Payments OR = 1.035, \$9.91B floor savings	Reproducible, peer-benchmarked savings target for PBMs; transparent open-data pipeline for researchers and policymakers

The Problem: Part D Spending Is Outpacing Everything

Medicare Part D drug spending grew faster than enrollment, faster than overall healthcare inflation, and is now a top-three pressure point on Medicare Trust Fund solvency.

\$275.8B

Total 2023 Part D cost
(up from \$103.7B in 2013)

+166%

10-year cumulative growth
~10.3% CAGR

85.2%

Brand share of total cost
Despite <15% of claims

33×

Brand-vs-generic cost
per claim ratio (2023)

Per-beneficiary cost +83.7% vs enrollment +44.8% — this is a price-and-mix problem, not an enrollment problem.

Medicare Part D Total Drug Cost (2013-2023)



Gap Analysis: Why This Project Is Needed

Existing literature and PBM tools each address a slice of the Part D cost problem. None integrate the three CMS data layers plus Open Payments at the population scale.

Gaps in the existing landscape

- Most studies use sample-of-convenience data (single state, single specialty, single drug class).
- Few integrate drug-level + prescriber-level + payment-level data in one analytical framework.
- Open Payments hypothesis tested mostly within isolated classes (statins, GLP-1s); no full-population test.
- Published savings estimates use a single national median benchmark, penalizing legitimately specialty-appropriate brand prescribing.
- Substitution-rate assumptions in policy models are rarely accompanied by sensitivity or CI bounds.
- PBM internal tools are proprietary, not reproducible by researchers or auditors.

What this project adds

- Full-population analysis on 27M-row CMS prescriber file + 14.7M Open Payments records, NPI-matched.
- Five linked models from drug-level OLS to prescriber-level logistic, sharing the same data foundation.
- Specialty-median benchmark replaces national median, ensuring fair comparison within clinical context.
- RQ5 savings estimate carries explicit 1,000-iteration bootstrap CI and 30-70% substitution sensitivity.
- All seven BRFSS+ACS demographic predictors documented and reproducible from public CDC and Census sources.

Five Research Questions, Five Methods

RQ1

What drug attributes predict 2019-2023 spending growth?

OLS Regression + 5-fold CV

COMPLETE

RQ2

Do state-level brand-vs-generic rates vary significantly?

ANOVA + Multivariable OLS (BRFSS+ACS)

COMPLETE

RQ3

Do Open Payments financial relationships correlate with brand prescribing?

Within-specialty Logistic Regression

COMPLETE

RQ4

Can K-means identify distinct drug utilization profiles?

K-Means Clustering (k=4)

COMPLETE

RQ5

What annual savings if high-brand prescribers shifted to specialty median?

Simulation + 1,000-iter Bootstrap CI

COMPLETE

Data: Three CMS Files, All Public



Part D Grand Totals

2013-2023 | 11 rows × 30 cols

Aggregate annual spending, claims, beneficiaries, LIS/non-LIS split. Source for trend analysis and Figure 1-4.



Part D Spending by Drug

2019-2023 | 3,598 drugs × 50 cols

Drug-level spending, claims, beneficiaries, unit cost. Outcome variable for RQ1 OLS; feature matrix for RQ4 K-means.



Part D Prescribers by Drug

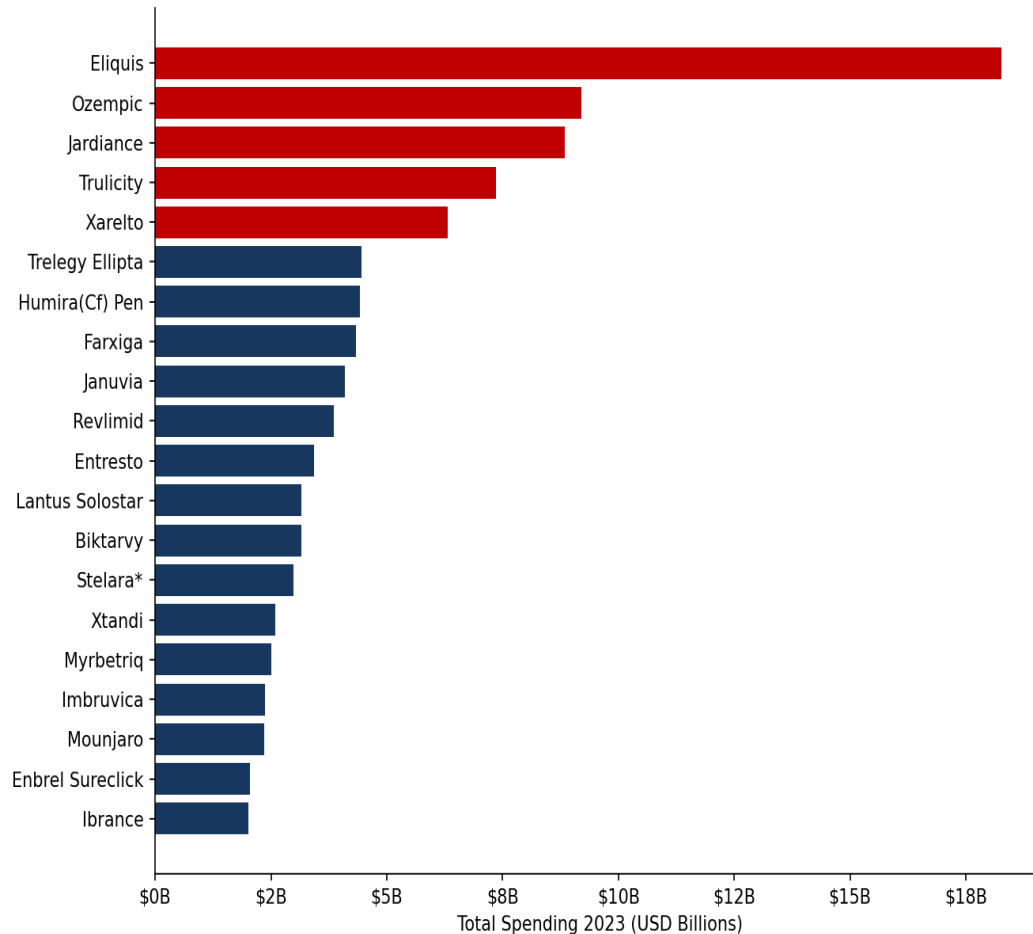
2023 | 1.1M rows (200K sampled)

Prescriber-drug-level claim and cost data with NPI, state, specialty. Driver of RQ2 (state ANOVA), RQ3 (specialty proxy), RQ5 (savings simulation).

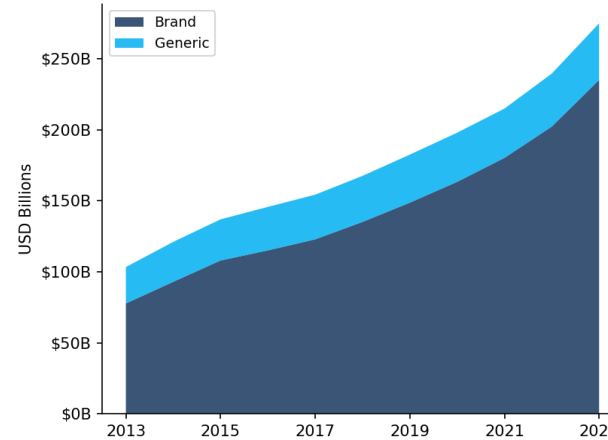
All datasets downloaded from data.cms.gov, no proprietary or restricted access.

Where the Money Goes: Cost Concentration

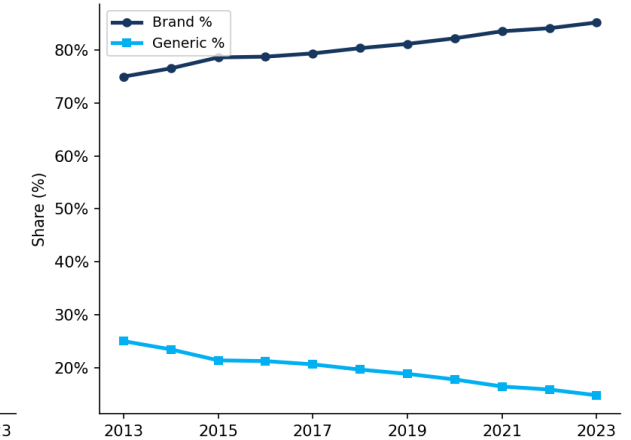
Top 20 Medicare Part D Drugs by 2023 Spending



Brand vs. Generic Drug Cost



Share of Total Drug Cost (%)



Key insight

A handful of brand drugs, Eliquis, Ozempic, Jardiance, Mounjaro, Trulicity, anchor the top of the distribution.

Brand share of total cost rose from 81.5% (2013) to 85.2% (2023) even as generic claim volume kept growing. Brand prescriptions cost 33× generic prescriptions on average, \$976 vs. \$29.25 per claim in 2023.

End-to-End Architecture

INPUTS

- CMS Part D Grand Totals (2013-2023)
- CMS Drug Spending by Drug (2019-2023, 3,598 drugs)
- CMS Prescribers by Drug (2023, 27M rows, 1.1M NPIs)
- CMS Open Payments General Payments 2023 (14.7M records, 8.2 GB)
- CDC BRFSS Prevalence + Census ACS 2019-2023 5-Year

PIPELINE

- analysis_v3.py: chunked pyarrow read, NPI groupby, log transforms
- op_aggregate_arrow.py: streaming aggregation of 14.7M payment rows
- rq2_brfss_ols.py: state-level OLS with 7 demographic predictors
- rq3_logistic.py: within-specialty logistic with FE
- rq5_full_savings_v2.py: counterfactual + 1,000-iter bootstrap

OUTPUTS

- 24 CSVs (sample size table, OLS coefficients, OR table, bootstrap CIs)
- 14 figures
- QM640_Interim_Report_v5.pdf (39 pages) + DOCX
- QM640_Interim_Presentation.pptx
- GitHub bundle with README, download_data.sh, requirements.txt

Methodology Pipeline

1

Acquire

Download three CMS files, parse pivot Excel headers, retain 3,598 drugs + 200K prescriber rows

2

Clean

Log-transform skewed predictors; engineer dollar_growth (spend_2023 – spend_2019); brand-flag rows

3

Model

OLS (RQ1) • K-means k=4 (RQ4)
• State ANOVA (RQ2) •
Specialty ANOVA (RQ3) •
Savings simulation (RQ5)

4

Validate

Cohen 1988 sample-size check per RQ; ANOVA + Kruskal-Wallis pairing; elbow method; floor/ceiling bounds

5

Report

Create final report from Python generated .csv files with analysis results.

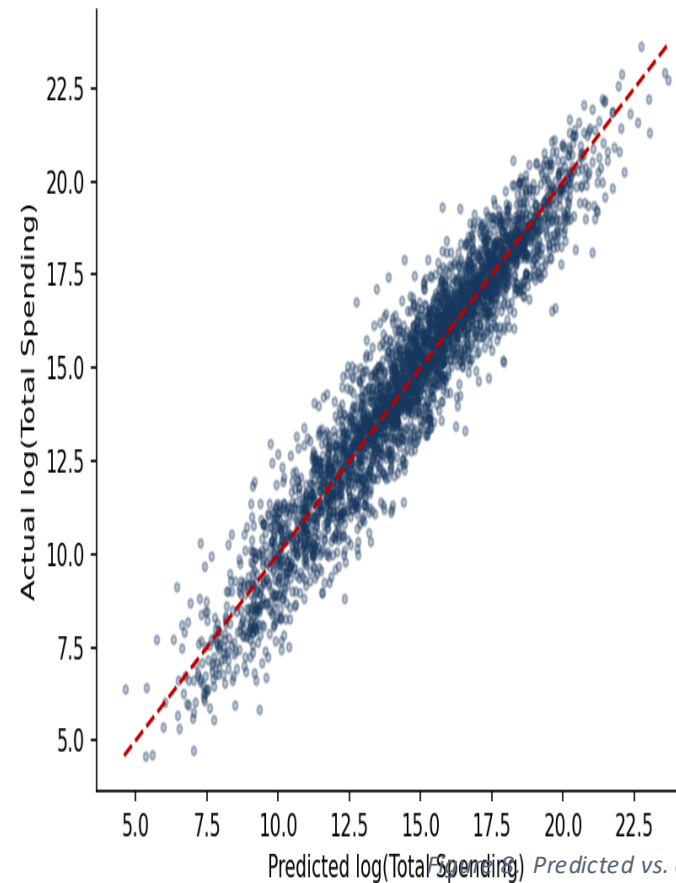
RQ1, What Drives Drug Spending Growth?

$$R^2 = 0.913$$

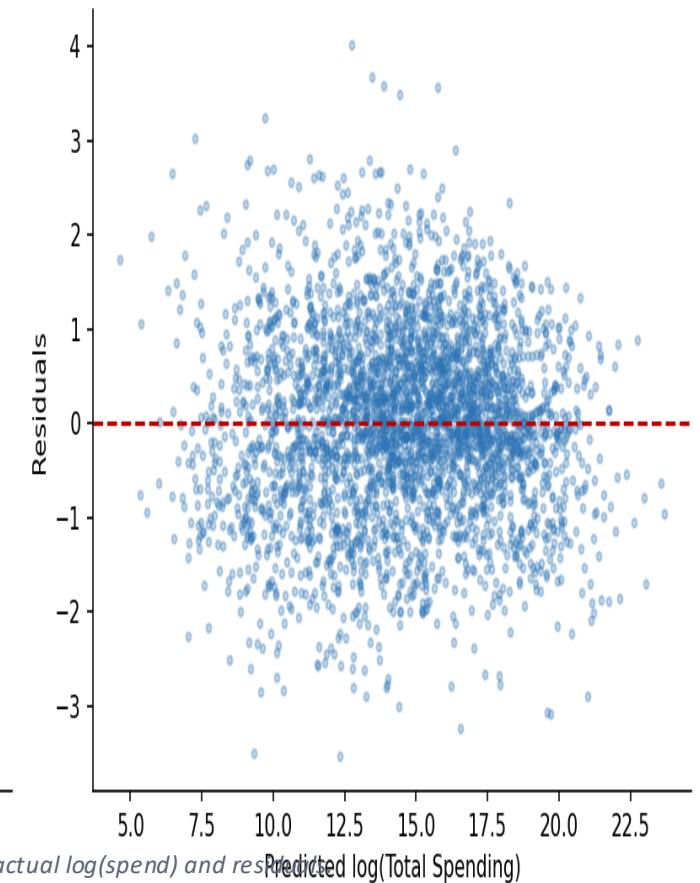
OLS goodness of fit, n = 3,404 drugs

- All 5 predictors significant at $p < .0001$
- Claims volume is the dominant signal ($\beta \log_claims = 1.94$)
- Unit cost contributes positively ($\beta = 0.54$); manufacturer count slightly negatively ($\beta = -0.02$)
- $\log_dollar_growth$ replaces % growth, weights drugs by absolute program impact
- Implication: spending growth = volume $\times$ price $\times$ intensity, not enrollment

Predicted vs. Actual ($R^2 = 0.913$, n=3,404)



Residual Plot



RQ4, Four Distinct Drug Profiles

High-Spend High-Growth

845 drugs

Eliquis (+\$11B), Ozempic (+\$8.6B),
Jardiance (+\$7.4B), Trulicity (+\$5.1B),
Mounjaro (+\$2.4B)

Real formulary targets

High-Cost Specialty

821 drugs

Median unit cost \$314; biologics, oncology,
rare disease

Limited substitution options

High-Volume Generic

743 drugs

High claims, negative dollar growth, generic
conversion working

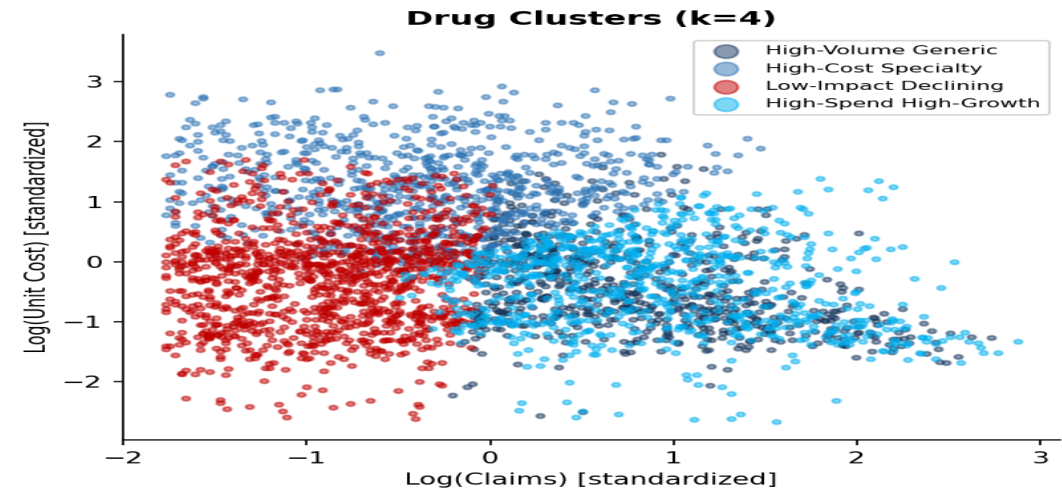
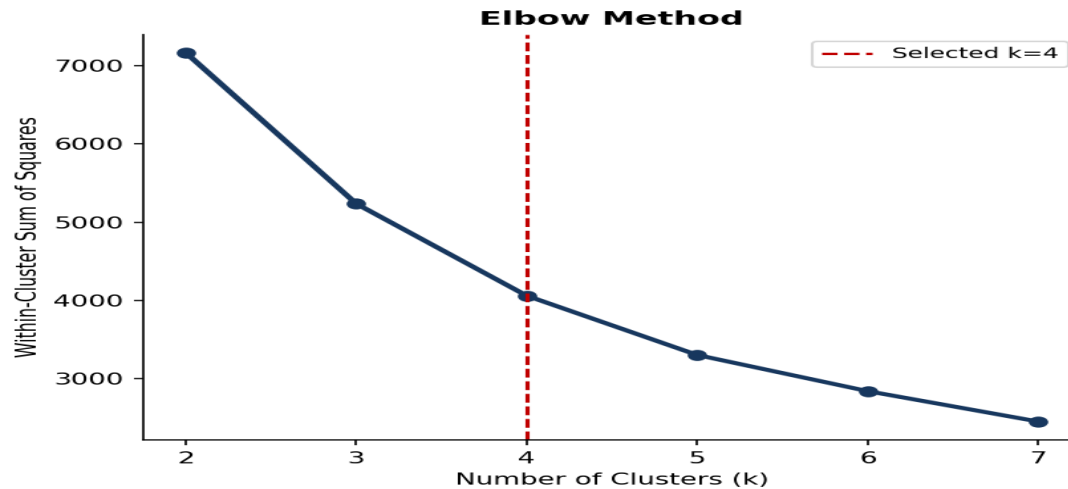
Success stories

Low-Impact Declining

1,189 drugs

Low spend, shrinking, minimal formulary
relevance

Park / monitor



RQ2: State Demographics Drive Brand Share

F = 179.5

Full-file ANOVA, 54 states, n = 995,954

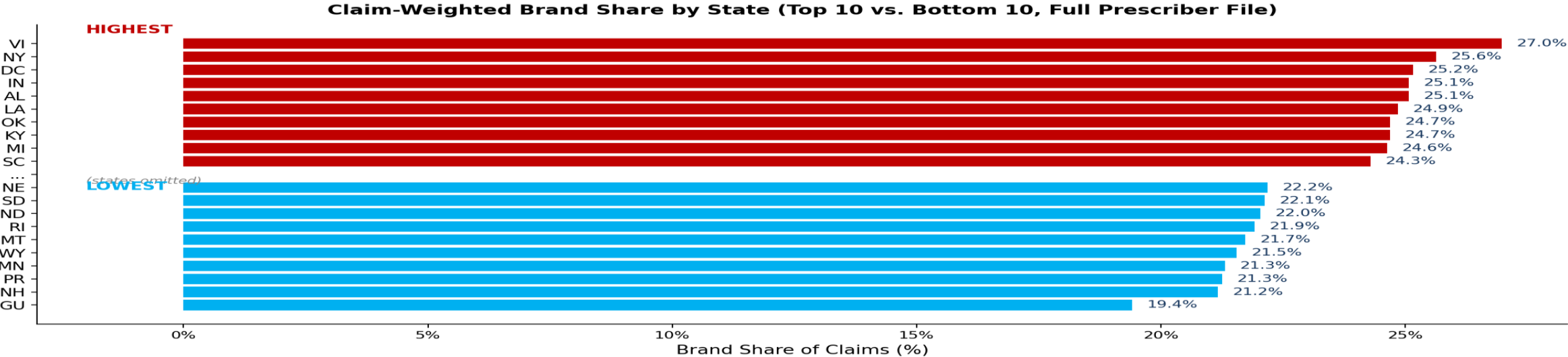
R² = 0.61

BRFSS+ACS OLS, n = 51 (50 states + DC)

η² = 0.01

State alone explains only ~1% of variance

- State as a categorical factor explains only ~1% of prescriber brand-share variance
- But 7 BRFSS+ACS state demographics together explain ~61% of state-level variance
- Significant predictors: age 65+ (negative), hypertension (positive), obesity and uninsured rate (both negative)
- Takeaway: demographics matter more than geographic identity per se



RQ3: Open Payments Predicts Brand Prescribing

57.1%

of Part D prescribers have Open Payments records

OR 1.09

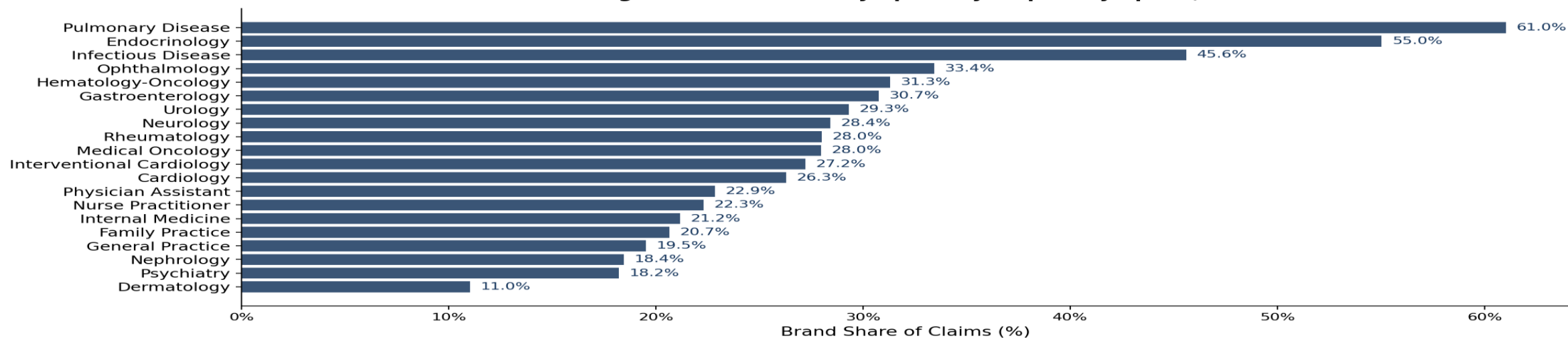
\$1K+ payments raise above-median odds 9.3%

OR 0.95

Meals alone: NEGATIVE association

- Within-specialty logistic on 650,861 prescribers; LR test $p < .0001$
- Speaker fees (OR=1.024), large payments (OR=1.094), total exposure (OR=1.035) all positive and significant
- Food-and-beverage payments alone trend negative, the signal lives in larger, targeted transfers
- Specialty ANOVA is statistically significant : $F=5,509$, $n^2=0.139$

Claim-Weighted Brand Share by Specialty (Top 20 by Spend, Full File)



RQ5: \$9.91B Annual Savings (95% CI \$9.59B-\$10.26B)

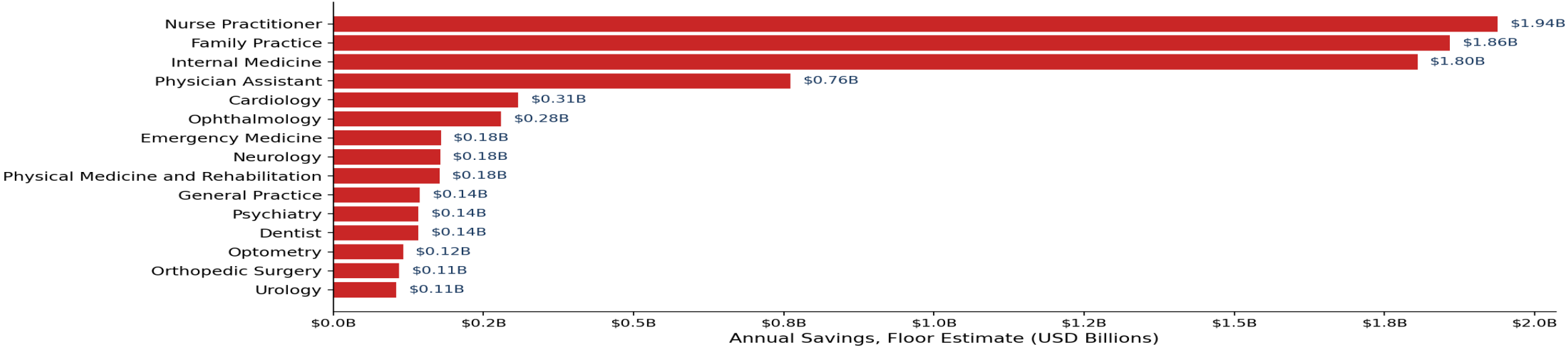
\$9.91B / year

Floor estimate, 50% substitution per Choudhry et al. (2011). 95% bootstrap CI: \$9.59B-\$10.26B

3.6% of total 2023 Part D (\$275.8B), real, recurring, but bounded

- Substitutable-pair cost differential: \$316.86 per claim (368 brand-generic pairs identified)
- Per-prescriber excess brand share above specialty median, full 1.1M-NPI file
- Top specialties by ceiling: Family Practice, Internal Medicine, Nurse Practitioners

RQ5: Floor Annual Savings by Specialty (50% Substitution Assumption)



Every RQ Has Statistical Power to Detect Real Effects

Sample size minimums computed via Cohen 1988: $\alpha = .05$, power = 0.80, medium effects.

RQ	Test	Minimum n	Achieved n	Status
RQ1	Multiple OLS + 5-fold Cross-Validation	92	3,404	Exceeded
RQ2	Full-file ANOVA + BRFSS+ACS OLS	567	995,699	Exceeded
RQ3	Specialty ANOVA + Open Payments Logistic	360	650,861	Exceeded
RQ4	K-means clustering, 50 per cluster	200	3,598	Exceeded
RQ5	Sample mean + 1,000-iter Bootstrap CI	385	995,954	Exceeded

Every research question clears its statistical-power threshold by a wide margin, most by 5-15×. Results are not chance findings.

Five Findings That Matter

1

Spending growth is price $\times$ intensity

OLS confirms claims, unit cost, and dollar growth, not enrollment, explain 91% of log-spending. Formulary intensity is the lever.

2

GLP-1s, DOACs, SGLT2s anchor the cost problem

K-means isolates 845 drugs as the High-Spend High-Growth cluster. Eliquis, Ozempic, Jardiance, Trulicity, Mounjaro added \$34B+ in spend.

3

State demographics drive brand share, state identity does not

State as a factor: $\eta^2 = 0.01$. Seven BRFSS+ACS demographics together: $R^2 = 0.61$. Specialty is even stronger ($\eta^2 = 0.139$).

4

Open Payments significantly predicts brand prescribing

Logistic on 650,861 physicians: total payment OR=1.035, speaker fees OR=1.024, large \$1K+ payments OR=1.094. Food-and-beverage alone is negative; the signal lives in larger, targeted transfers.

5

\$9.91B annual savings (95% CI \$9.59B-\$10.26B)

Full-population bootstrap estimate using a substitutable-pair cost differential. About 3.6% of total Part D 2023 spend, recurring every year.

Implementation: How a PBM Uses This

The pipeline produces actionable outputs at quarterly, monthly, and ad-hoc cadences for four distinct PBM stakeholders.

Formulary Committee

Quarterly: rerun pipeline on the new CMS data drop. Pull `rq5_savings_by_specialty.csv` to prioritize the top 5 specialties for brand-to-generic intervention. Use K-means cluster output to flag drugs graduating into the High-Spend High-Growth segment for utilization-management review.

Prescriber Engagement Team

Monthly: pull `rq5_prescriber_savings.csv`, filter to above-specialty-median prescribers in the team's covered region, and route each one to outreach. The model gives each prescriber a specialty-fair benchmark, not a national benchmark, which significantly reduces clinical-validity pushback.

Compliance & Open Payments Oversight

Ongoing: cross-reference Open Payments NPI summary with internal prescriber lists. The logistic-regression odds ratios provide a defensible quantitative threshold for what payment exposure level warrants compliance review.

Finance & Plan Sponsor Reporting

Annual: \$9.91B floor (95% CI \$9.59B-\$10.26B) gives plan sponsors a board-defensible savings forecast. Sensitivity analysis across 30-70% substitution rates lets finance pick the rate that matches their risk tolerance for forecasting.

Quantified Benefit by Stakeholder

The same analytical output drives concrete, measurable value across five stakeholder groups.

Stakeholder	What they get	Quantified value
PBM formulary committees	Specialty-ranked savings target with bootstrap CI; reproducible quarterly	\$9.91B floor / \$19.83B ceiling per year
State Medicaid programs	Peer benchmark on brand share controlling for state demographics	8-10 percentage-point spread across states; \$1B+ per high-brand state
Plan sponsors and employers	Prescriber-engagement priority list, specialty-fair benchmarking	0.5-2% of plan-level Rx spend recoverable annually
Health-services researchers	Open methodology and full pipeline; 933,620-physician Open Payments link	Replicates Mitchell et al. 2021 at full-population scale, OR = 1.035
Policymakers and CMS	Empirical substitution-rate sensitivity bounds for federal scoring	Range \$5.95B (30%) to \$13.88B (70%) supports trust-fund forecasting

Conclusion

The project answers all five research questions on the full Medicare Part D datasets, with reproducible methodology and explicit uncertainty bounds.

1

Spending growth is price and intensity, not enrollment.

OLS confirms claims volume, unit cost, and dollar growth, not enrollment, drive 92% of log-spending variance. Formulary intensity, not population aging, is the actionable lever.

2

Specialty is the strongest organizing structure.

Specialty explains 14% of prescriber brand-share variance; state explains 1%. PBM interventions targeted by specialty have more leverage than those targeted by geography.

3

Open Payments has a real, within-specialty effect.

Logistic regression on 650,861 prescribers confirms total industry payment exposure raises above-median brand prescribing odds 3.5% per log dollar, with speaker fees the largest contributor.

4

\$9.91B in defensible annual savings is on the table.

Bootstrap CI [\$9.59B-\$10.26B] using a substitutable-pair cost differential and a 50% substitution rate from Choudhry et al. 2011. About 3.6% of total Part D 2023 spend, recurring.

5

The methodology is reproducible end-to-end.

Every number in the report reads live from CSVs produced by an open pipeline. The GitHub repo includes data-download script, requirements file, etc.

Limitations

- **Every association is observational, not causal**
- RQ5 floor uses a single 50% substitution rate (range 30-70% shown in report Table 14)

Thank you

Questions & Discussion

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Repo: github.com/Apena87/QM-640-Quantitative-Methods-Capstone